

Chain Poker Genesis

Pot Limit Omaha Rules Engine

Official Technical Document v1.0

1. Purpose

The Pot Limit Omaha Rules Engine is a specialized subsystem of Chain Poker Genesis responsible for implementing all official rules of the Pot Limit Omaha (PLO) poker variant.

Its purpose is to validate every action performed during a game and ensure that all operations strictly comply with the official game rules.

This engine operates independently from all other protocol components and is selected exclusively during table creation.

Once a table has been created, the selected rules engine becomes permanently locked and cannot be modified.

2. Objectives

The engine shall guarantee:

- Full compliance with the official Pot Limit Omaha rules.
 - Deterministic validation of every game action.
 - Compatibility with the decentralized consensus system.
 - Identical behavior across all network nodes.
 - Inability to modify the rules during an active game.
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3. Scope

The engine is responsible for:

- Dealing four private hole cards to each player.
 - Validating Pot Limit betting rules.
 - Managing the betting round sequence.
 - Enforcing the mandatory use of exactly two private cards and three community cards to form the final poker hand.
 - Determining the winning hand.
 - Resolving tied hands.
 - Performing complete validation of the game state.
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4. Engine Selection

The Pot Limit Omaha Rules Engine can only be selected during the creation of a new table.

Once the game begins:

- The selected engine cannot be replaced.
 - Dynamic rule modifications are not permitted.
 - Every hand played at the table follows the exact same rule set.
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5. Rules Implemented

The engine enforces:

- Four private hole cards per player.
 - Five community cards.
 - Mandatory use of exactly two private cards.
 - Mandatory use of exactly three community cards.
 - Pot Limit betting structure.
 - Official order of play.
 - Official poker hand rankings.
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6. Bet Validation

The engine automatically calculates:

- Current pot size.
- Minimum legal bet.
- Maximum legal bet.

- Legal raise amounts.
- Pot Limit restrictions.

No player may place a bet outside these validated parameters.

7. Hand Validation

The engine automatically verifies:

- Correct number of cards used.
 - Legal hand combinations.
 - Hand ranking.
 - Comparison against all active players.
 - Deterministic winner determination.
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8. Integration with Chain Poker Genesis

The Pot Limit Omaha Rules Engine operates in conjunction with:

- Shuffle Engine.
- Dealer Engine.
- Conflict Resolution Engine.
- Reputation Engine.
- Consensus Layer.
- Communication System.
- Smart Contracts.

Each component maintains completely independent responsibilities.

9. Security

The engine guarantees:

- Rule immutability.
- Cryptographic validation of player actions.
- Consensus across participating nodes.
- Complete reproducibility of every hand.
- Inability to alter the rule set during gameplay.

10. Extensibility

The architecture allows additional rule engines to be incorporated without modifying the protocol core.

Examples include:

- Texas Hold'em Rules Engine.
- Omaha Hi-Lo Rules Engine.
- Five Card Omaha Rules Engine.
- Short Deck Rules Engine.
- Seven Card Stud Rules Engine.
- Draw Poker Rules Engine.

Each rules engine operates as an independent module while sharing the common Chain Poker Genesis infrastructure.

11. Conclusion

The Pot Limit Omaha Rules Engine is the protocol component responsible for deterministically, verifiably, and immutably enforcing all official Pot Limit Omaha rules within Chain Poker Genesis. Its modular architecture allows the game variant to be selected exclusively during table creation, preserving protocol integrity while enabling future poker variants to be added without altering the system's core architecture.

Chain Poker Genesis

Future Vision: Poker Game Variant Engines

Technical Strategy Document

1. Vision

The long-term vision of Chain Poker Genesis extends beyond implementing a single poker variant. The protocol is designed as a modular platform capable of supporting virtually every recognized poker game through specialized Rules Engines.

Rather than modifying the protocol core for each new game, Chain Poker Genesis introduces independent Game Variant Engines that encapsulate the complete rule set of a specific poker modality.

This architecture enables unlimited protocol evolution while preserving security, consensus, and compatibility across the network.

2. Modular Architecture

Every poker variant is implemented as an independent Rules Engine.

During table creation, the table creator selects the desired engine. Once selected, the engine becomes immutable for the lifetime of the table.

All remaining protocol components—including the Shuffle Engine, Dealer Engine, Reputation Engine, Conflict Resolution Engine, Communication System, Consensus Layer, and Smart Contracts—remain unchanged regardless of the selected poker variant.

3. Advantages

The modular engine architecture provides:

- Infinite protocol extensibility.
 - Independent development of new poker variants.
 - Simplified maintenance.
 - Complete backward compatibility.
 - Deterministic consensus.
 - Rule immutability.
 - Community-driven expansion.
 - Standardized validation across every supported game.
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4. Current and Future Poker Variant Engines

The architecture is intended to support engines for virtually every recognized poker modality, including but not limited to:

Community Card Poker

- Texas Hold'em
 - No Limit Texas Hold'em
 - Pot Limit Texas Hold'em
 - Fixed Limit Hold'em
 - Omaha Pot Limit (PLO)
 - Omaha No Limit
 - Omaha Fixed Limit
 - Omaha Hi-Lo (Eight or Better)
 - Five Card Omaha
 - Five Card Omaha Hi-Lo
 - Six Card Omaha
 - Courchevel
 - Courchevel Hi-Lo
 - Fusion Poker
 - Irish Poker
 - Double Board Omaha
 - Double Flop Omaha
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Stud Poker

- Five Card Stud
 - Seven Card Stud
 - Seven Card Stud Hi-Lo
 - Razz
 - Mexican Stud
 - Caribbean Stud Poker
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Draw Poker

- Five Card Draw
- Triple Draw Lowball
- Deuce-to-Seven Triple Draw

- Ace-to-Five Lowball
 - Badugi
 - Badacey
 - Badeucey
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Mixed Games

- HORSE
 - HOSE
 - HOE
 - 8-Game Mix
 - Dealer's Choice
 - Mixed Omaha Variants
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Short Deck Variants

- Short Deck Hold'em (6+ Hold'em)
 - Short Deck Omaha
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Chinese Poker

- Open-Face Chinese Poker
 - Open-Face Pineapple
 - Chinese Poker
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Pineapple Variants

- Pineapple
 - Crazy Pineapple
 - Tahoe Poker
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High-Low Variants

- Omaha Hi-Lo
- Stud Hi-Lo

- Split Pot Games
 - Lowball Variants
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Casino Poker Variants

- Three Card Poker
 - Four Card Poker
 - Let It Ride
 - Ultimate Texas Hold'em
 - Pai Gow Poker
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Experimental and Community Variants

The modular architecture also enables future community-developed poker variants to be integrated without modifying the protocol core, provided they comply with the Chain Poker Genesis Rules Engine specification.

5. Future Expansion

The architecture is intentionally designed to allow continuous growth. New Rules Engines can be developed, audited, certified, and deployed independently while maintaining full interoperability with the existing protocol.

This approach transforms Chain Poker Genesis into a universal decentralized poker infrastructure rather than a protocol limited to a single game.

6. Conclusion

The future of Chain Poker Genesis lies in the continuous development of specialized Poker Game Variant Engines. By separating game rules from the protocol infrastructure, the platform achieves scalability, flexibility, and long-term sustainability while preserving the security, determinism, and decentralization that define the Chain Poker Genesis ecosystem.